

CS5487: Machine Learning: Principles and Practice

Assignment 1

Semester A, 2026/27

Name: _____

Student ID: _____

Total: 100 marks

- There are eight questions.
- Questions 1–4 are calculation problems worth 50 marks in total.
- Questions 5–8 are derivation problems worth 50 marks in total.
- Show the essential reasoning leading to each answer.
- Numerical answers without supporting work may receive limited credit.
- A calculator may be used for numerical evaluation.

Question	1	2	3	4	5	6	7	8	Total
Marks	10	10	10	20	10	10	10	20	100

Question 1: Gaussian Geometry**[10 marks]**

A two-dimensional observation is modeled as

$$\mathbf{X} \sim \mathcal{N} \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \right).$$

Consider

$$\mathbf{x}_A = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

- (a) Compute the inverse covariance matrix. [2 marks]
- (b) Determine which observation has the larger Gaussian density under the model. Your calculation must use the geometry of the multivariate Gaussian rather than Euclidean distance alone. [5 marks]
- (c) Compute

$$\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)}.$$

You may leave the answer in exponential form.

[3 marks]

Question 2: Maximum-Likelihood Linear Regression [10 marks]

Four observations are

$$(-1, 0), \quad (0, 1), \quad (1, 2), \quad (2, 2),$$

where each pair is (x_i, y_i) .

Assume

$$y_i = w_0 + w_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2),$$

where w_0, w_1 , and σ^2 are all unknown.

- (a) Find the maximum-likelihood estimates of w_0 and w_1 . Show the system of equations you solve. [5 marks]
- (b) Using your fitted regression model, find the maximum-likelihood estimate of σ^2 . [3 marks]
- (c) Compute the predicted response at $x_* = 3$. [2 marks]

Question 3: Bayesian Inference over Discrete Models [10 marks]

An unknown Bernoulli success probability π is known to take one of only three possible values:

$$\pi \in \{0.2, 0.5, 0.8\}.$$

Before observing data, the prior probabilities are

$$P(\pi = 0.2) = 0.2, \quad P(\pi = 0.5) = 0.5, \quad P(\pi = 0.8) = 0.3.$$

You then observe the four outcomes

$$1, 1, 1, 0.$$

- (a) Compute the posterior probability of each possible value of π . [5 marks]
- (b) Determine the MAP estimate of π . [2 marks]
- (c) Using the full posterior distribution rather than the MAP estimate, compute the posterior predictive probability

$$P(x_* = 1 \mid \mathcal{D})$$

for the next Bernoulli trial.

[3 marks]

Question 4: One Complete EM Iteration**[20 marks]**

Consider the observations

$$x_1 = 0, \quad x_2 = 2, \quad x_3 = 4.$$

A two-component one-dimensional Gaussian mixture is initialized with

$$\pi_1^{(0)} = 0.6, \quad \pi_2^{(0)} = 0.4,$$

$$\mu_1^{(0)} = 0, \quad \mu_2^{(0)} = 4,$$

and

$$(\sigma_1^2)^{(0)} = (\sigma_2^2)^{(0)} = 2.$$

Perform one complete EM iteration. Do not assume hard cluster assignments.

- (a) Carry out the E-step and compute all responsibilities

$$\gamma_{ik}^{(0)}, \quad i = 1, 2, 3, \quad k = 1, 2.$$

Report answers to four decimal places.

[8 marks]

- (b) Carry out the relevant M-step calculations to obtain

$$\pi_1^{(1)}, \quad \pi_2^{(1)}, \quad \mu_1^{(1)}, \quad \mu_2^{(1)}.$$

[7 marks]

- (c) Complete the M-step by computing

$$(\sigma_1^2)^{(1)} \quad \text{and} \quad (\sigma_2^2)^{(1)}.$$

[5 marks]

Use the Gaussian density itself when computing the E-step; no responsibility values are provided.

Question 5: Variance of a Linear Projection**[10 marks]**

Let

$$\mathbf{X} \in \mathbb{R}^d, \quad \mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{X}) = \boldsymbol{\Sigma}.$$

For fixed $\mathbf{a} \in \mathbb{R}^d$ and $b \in \mathbb{R}$, define

$$Y = \mathbf{a}^\top \mathbf{X} + b.$$

Starting from the definitions of expectation and variance:

- (a) Derive the mean of Y . [3 marks]
- (b) Derive the variance of Y and express the final result using $\mathbf{a}$ and $\boldsymbol{\Sigma}$. [7 marks]

Question 6: MLE with Unequal Observation Variances [10 marks]

Suppose

$$x_i \sim \mathcal{N}(\mu, \sigma_i^2), \quad i = 1, \dots, n,$$

independently, where the variances

$$\sigma_1^2, \dots, \sigma_n^2$$

are known but need not be equal. The common mean μ is unknown.

(a) Write the log-likelihood of μ up to terms that do not depend on μ . [3 marks]

(b) Differentiate the log-likelihood and derive the closed-form maximum-likelihood estimator of μ . [5 marks]

(c) Simplify your result when

$$\sigma_1^2 = \dots = \sigma_n^2.$$

[2 marks]

Question 7: MAP Estimation with a Laplace Prior [10 marks]

Suppose

$$x_i \mid w \stackrel{\text{iid}}{\sim} \mathcal{N}(w, \sigma^2), \quad i = 1, \dots, n,$$

where σ^2 is known.

Instead of a Gaussian prior, assume

$$p(w) \propto \exp(-\lambda|w|), \quad \lambda > 0.$$

Let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- (a) Derive the negative log-posterior objective for w , ignoring terms independent of w . [3 marks]
(b) Solve the MAP optimization separately for the cases

$$w > 0 \quad \text{and} \quad w < 0.$$

[4 marks]

- (c) Include the possibility $w = 0$ and derive one closed-form expression for

$$\hat{w}_{\text{MAP}}$$

as a function of $\bar{x}$, λ , σ^2 , and n .

[3 marks]

Question 8: From Gaussian Mixtures to K-Means [20 marks]

Consider a Gaussian mixture model with K components satisfying

$$\pi_k = \frac{1}{K}, \quad \Sigma_k = \sigma^2 \mathbf{I}$$

for every component k .

The component means are

$$\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K.$$

(a) Starting from Bayes' rule and the Gaussian density, derive an expression for the responsibility

$$\gamma_{ik} = P(z_i = k \mid \mathbf{x}_i)$$

in terms of the squared Euclidean distances

$$\|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2.$$

[5 marks]

(b) For two components a and b , derive the ratio

$$\frac{\gamma_{ia}}{\gamma_{ib}}$$

in terms of

$$\|\mathbf{x}_i - \boldsymbol{\mu}_a\|_2^2 \quad \text{and} \quad \|\mathbf{x}_i - \boldsymbol{\mu}_b\|_2^2.$$

[5 marks]

(c) Assume component k^* is the unique nearest centroid to $\mathbf{x}_i$. Using your result from part (b), determine the limits of

$$\gamma_{ik^*} \quad \text{and} \quad \gamma_{ij}, \quad j \neq k^*,$$

as

$$\sigma^2 \rightarrow 0.$$

[5 marks]

(d) Now suppose these limiting hard assignments are written as

$$r_{ik} \in \{0, 1\}.$$

Starting from

$$J_k(\boldsymbol{\mu}_k) = \sum_i r_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2,$$

derive the centroid value that minimizes J_k .

[5 marks]